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Targeting TAF1 with BAY-299 induces antitumor immunity in triple-negative breast cancer

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ABSTRACT

Triple-negative breast cancer (TNBC) is a heterogeneous breast cancer subtype with poor prognoses and limited therapeutic options. The TATA-box binding protein associated factor 1 (TAF1) is an essential protein involved in the transcriptional regulation of cancer development and progress. However, the therapeutic potential and underlying mechanism of targeting TAF1 in TNBC remain unknown. Here, using chemical probe BAY-299, we identify that TAF1 inhibition leads to the induction of endogenous retrovirus (ERVs) expression and double-stranded RNA (dsRNA) formation, resulting in the activation of interferon responses and cell growth suppression in a subset of TNBC, resembling anti-viral mimicry effect. This correlation between *TAF1* and interferon signature was validated in three independent breast cancer patient datasets. Furthermore, we observe heterogeneous responses to TAF1 inhibition across a set of TNBC cell lines. By integrating transcriptome and proteome data, we demonstrate that high levels of proliferating cell nuclear antigen (PCNA) protein serve as a predictive biomarker associated with suppressive tumor immune responses in various cancers, which may limit the efficiency of TAF1 inhibition.

1. Introduction

Triple-negative breast cancer (TNBC) is the most aggressive and heterogeneous breast cancer subtype with poor prognosis, elevated metastasis, and significant mortality rates [1], accounting for 15–20% of all breast cancer cases [2]. TNBC patients are characterized by the lack of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2) expression, resulting in insensitivity to HER2-directed and/or endocrine therapies, which are the most commonly employed in breast cancer treatment [3]. Currently, the combination of surgery and chemotherapy followed by radiation therapy is still the mainstay of treatment for early and metastatic TNBC [4]. Therefore, the identification of novel therapeutic targets and the development of effective drugs for TNBC treatment are urgently needed.

The TATA-box binding protein associated factor 1 (TAF1) protein is an emerging epigenetic target for cancer therapy [5,6]. TAF1 is the largest subunit of transcription factor IID (TFIID) complex, bridging the recruitment of other TAF family proteins for RNA polymerase II-dependent transcription initiation in eukaryotic cells [7]. In addition, TAF1 protein contains a bromodomain, which endows it with histone acetyltransferase (HAT) activities and enables regulation of gene expression associated with cell cycle and apoptosis [8]. Specifically, TAF1 catalyzes the acetylation of histone 3 in the promotor regions of cell cycle associated genes, such as cyclin D1 and cyclin A [9]. Growing evidences suggests that TAF1 plays a crucial role in cancer development and progress. For example, TAF1 was reported to promote glioma tumorigenesis by binding to long non-coding RNA 319 (LIN00319) [10]. In non-small cell lung cancer (NSCLC), TAF1 induced cell epithelial-mesenchymal transition by transcriptionally activating TGF beta 1 [11]. TAF1 is also a co-activator of androgen receptor that directly binds and enhances AR transcriptional activity in prostate cancer [12]. Given these essential roles of TAF1 in tumor development,

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targeting TAF1 has emerged as a promising therapeutic strategy for cancer treatment [13].

However, the regulatory effect of TAF1 and its potential as a therapeutic target in TNBC remain unknown. In this study, using a chemical probe BAY-299, we uncovered the antitumor immunity of TAF1 inhibition in a subset of TNBC. Mechanistically, TAF1 inhibition stimulated the antiviral interferon (IFN) pathway by the induction of the endogenous retrovirus (ERV) expression and double-stranded RNA (dsRNA) formation. In parallel, BAY-299 treatment increased the expression of genes associated with chemo-attractants, such as *CXCL9, CXCL10* and *CCL5*, and antigen processing and presentation, such as *B2M* and *HLA-C*. These effects hold the potential to improve the immunotherapy by T cell recruitment and activation. Supporting these findings, the correlation between *TAF1* and IFN-alpha signature was validated in three independent public clinical datasets. In addition, analysis of transcriptome and proteome data revealed that the expression level of proliferating cell nuclear antigen (PCNA) protein was associated with the anti-tumor efficacy of BAY-299. *PCNA* was also found to be correlated with NKT cell infiltration across multiple cancers, highlighting its effect in the regulation of TNBC immunity.

2. Materials and methods

2.1. Cell culture

Human TNBC cell lines (HCC1395, MDA-MB-231, CAL-120, HCC3153, HCC1806, MDA-MB-468, HCC70, HCC1143, HCC38, Hs 578T, MB 157, BT-549) were cultured in RPMI 1640 (Gibco, C11875500BT) or DMEM (Gibco, C11995500BT) with 10% FBS (BIOIND, 04-001-01ACS) and 1% penicillin/streptomycin (Biosharp, BL505A).

2.2. Cell proliferation and colony formation assay

For cell proliferation assay, TNBC cell lines (500 cells/well) were plated on 384-well plates. Compounds were dissolved in DMSO and sequentially diluted from 10 μM to 0.02 μM. Confluency was obtained from an Incucyte SX5 live cell imaging device (Sartorius, Germany). The half maximal inhibitory concentration (IC_{50}) and area above the curve (AAC) were calculated using GraphPad software (v8.0.2).

For colony formation assay, MDA-MB-231 cells (1000 cells/well) were seeded in six-well plates. After 12 days, cells were fixed with 4% multigrade formaldehyde (Biosharp, BL539A) and stained with crystal violet solution (Beyotime, C0121) for 1 h.

2.3. Apoptosis assay

MDA-MB-231 cells (3000 cells/well) were seeded in 96-well plates and treated with DMSO or BAY-299 (1 μM) for 5 days. Annexin V reagent (Vazyme, A211-01) was added and analyzed with Incucyte.

2.4. Epigenetic targets dependency screening on breast cancer panel

Epigenetic targets dependency data of breast cell lines were obtained from CRISPRGeneDependency on the Cancer DepMap portal of the CCLE dataset (https://depmap.org/portal/download/all/). The dependency heatmap was performed with R package pheatmap (v1.0.12).

2.5. Small interfering RNA (siRNA)-mediated gene knockdown

SiRNA targeting *TAF1* were purchased from GENERAL BIOL (Anhui, China). Cells were transfected with 50 nmol/L siRNA using lipofectamine 2000 reagent (Invitrogen, 11668019). SiRNA sequence were summarized in Table 1.

2.6. RNA-sequencing analysis

Differential gene expression analysis was performed using the R package DESeq2 (v1.34). Gene set enrichment analysis (GSEA) was performed on a list of genes ranked from high to low DESeq2 estimated fold-change using the R package clusterprofiler (v4.6.0).

2.7. Inverted transposable elements analysis

The expression of inverted transposable elements was analyzed with RNA-seq data. Bowtie2 (v2.2.5) was used for mapping the data to GRCh38. Samtools (v1.10) was used for converting the.sam files to.bam files. RepEnrich2_subset was used to get uniquely and multi-mapped reads.

2.8. Immunofluorescence staining

Cells were fixed in 4% multigrade formaldehyde and blocked in 5% bovine serum albumin (Biosharp, 9048-46-8). J2 antibody (Scicons, 10010200, 1:500), anti-mouse IgG (Beyotime, A0428, 1:500) and DAPI (Beyotime, C1005) were added and incubated. Images were detected on a Sigle-Photon confocal microscope (NIKON, A1 HD25) and analyzed by ImageJ (v1.8.0).

2.9. RT-qPCR

Total RNA was isolated with the Total RNA Extraction Reagent kit (Vazyme, RC112-01). A cDNA Synthesis kit (Vazyme, R223-01) was used for synthesizing cDNA. Real-time quantitative PCRs were performed using SYBR (Vazyme, Q712-02) on LightCycler 96 Real-time PCR system.

2.10. Correlation between TAF1 and $CD8^+$ T cell infiltration

Online database TIMER2.0 [14] TIMER2.0 (cistrome.org) was applied to assess the correlation between *TAF1* and $CD8^+$ T cell infiltration.

2.11. Gene set enrichment analysis of cell lines

AAC was Pearson-correlated to the level of individual gene expression levels across cell lines. The resulting ranked Pearson correlation scores were used as an input for GSEA.

2.12. Proteomic analysis

Princess Margaret Cancer Center protein expression data was downloaded [15]. Pearson correlation coefficients were calculated to measure the correlation between AAC and individual protein expression levels across 12 TNBC cell lines.

Table 1
Primer sequences.

TAF1 (human)-siRNA	CCAAUCAACUGGAGAUAUATT UAUAUCUCCAGUUGAUUGGTT
control	UUCUCCGAACGUGUCACGUTT ACGUGACACGUUCGGAGAATT

3. Results

3.1. TAF1 is a therapeutic target in TNBC

To define the potential significance of TAF1 in TNBC, we first conducted a comparative analysis of TAF1 expression between tumor and normal samples in TCGA and GTEx datasets. As expected, *TAF1* mRNA level was upregulated in breast cancer tissues compared to the normal ones ($P = 0.0056$, Fig. 1a), suggesting the therapeutic potential of targeting TAF1 in TNBC treatment. Since a specific TAF1 inhibitor is not yet available, we used BAY-299, a dual inhibitor targeting both TAF1 and BRD1 (Synonyms: BRL or BRPF2) for further investigation. Treatment of the human TNBC MDA-MB-231 cell line with BAY-299 impaired cell growth in a dose dependent manner with an $IC_{50} = 1.375$ μM (Fig. 1b). This anti-proliferative effect was further validated in a colony-formation assay. We found that BAY-299 treatment for 12 days significantly inhibited colony formation of MDA-MB-231 (Fig. 1c). To determine whether TAF1 inhibition induced cell apoptosis, we stained cells treated with BAY-299 or vehicle control with apoptosis agent, Annexin V. A dose-dependent increase of apoptotic cells were observed in BAY-299 treated cells (Fig. 1d–e). To exclude the effect of *BRD1*, a member of the bromodomain and PHD finger family protein [16], on BAY-299 induced cell growth suppression, we analyzed the essentiality of *TAF1* and *BRD1* across 21 TNBC cell lines in public human Cancer Dependency Map dataset [17]. *BRD1* is much less essential for TNBC cell growth compared to *TAF1*, highlighting the dominate effect of *TAF1* inhibition on BAY-299 treatment (Fig. 1f). Taken together, these data supported that pharmacologically targeting TAF1 effectively limit TNBC progression.

3.2. TAF1 inhibition induces interferon response and enhances antigen processing and presentation

To understand the underlying mechanism of TAF1 inhibition in TNBC, MDA-MB-231 cells were treated with BAY-299 or vehicle control for 5 days, and harvested for RNA sequencing. Transcriptomics analysis identified significant changes in the expression of total 227 genes upon BAY-299 treatment, including 150 genes with increased expression and 77 genes with reduced expression ($P < 0.05$, Fig. 2a). Gene set enrichment analysis (GSEA) revealed that interferon alpha and gamma responses were the two most positively enriched pathways in BAY-299 treated cells, suggesting that TAF1 inhibition activated the expression of genes associated with innate immune responses (Fig. 2b). The upregulation of IFN-gamma upon TAF1 inhibition was further validated by qPCR measurement (Fig. 2d). RNA-sequencing and qPCR validation also demonstrated the upregulation of IFN-stimulated genes, such as *MX1*, *OAS1*, *OAS2*, *OASL*, and *ISG15*, indicating the induction of anti-

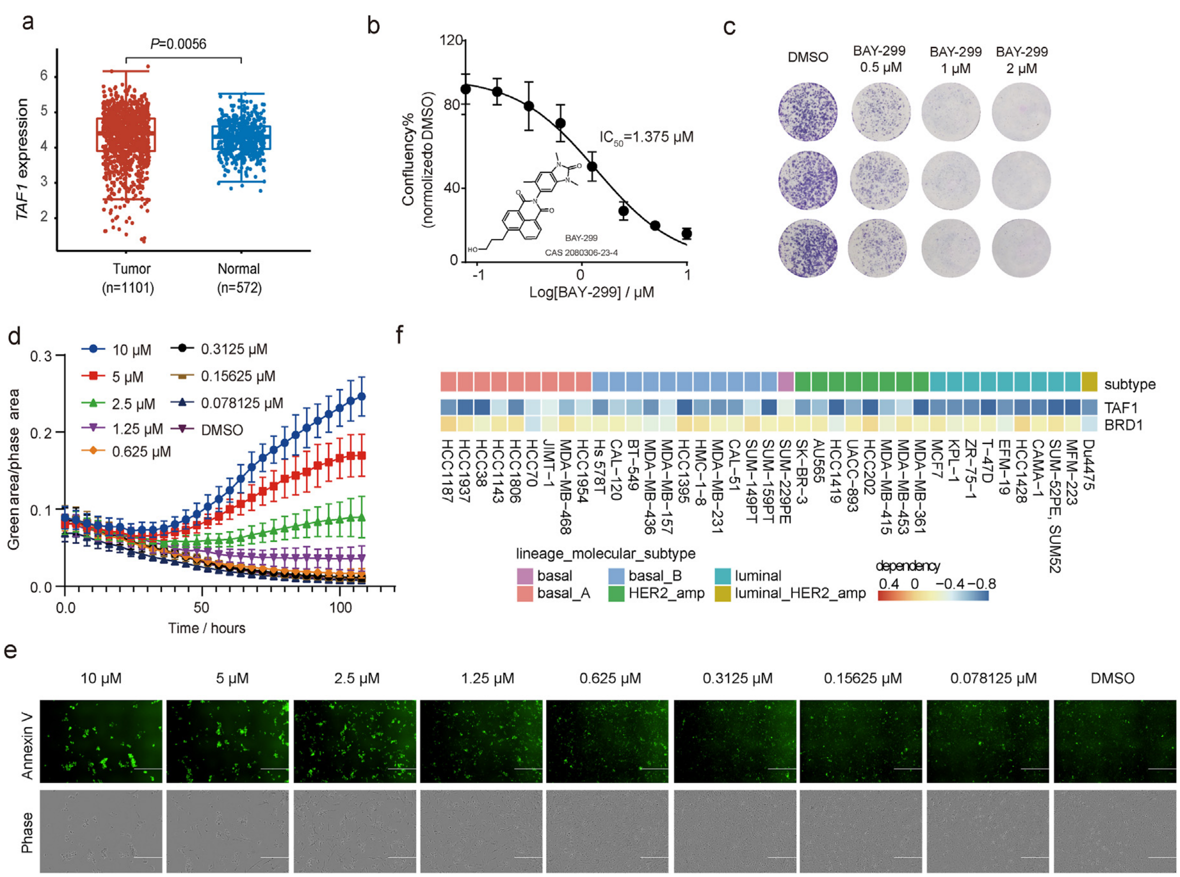

Fig. 1. TAF1 inhibition suppresses MDA-MB-231 cell growth. (a) Relative *TAF1* mRNA expression in breast cancer (n = 1101) and normal tissues (n = 572) from TCGA and GTEx dataset (Wilcoxon test); (b) Growth curves of MDA-MB-231 treated with BAY-299 or DMSO for 5 days. The structure of BAY-299 is shown as indicated. IC_{50} value for BAY-299 is shown on the right (n = 6, mean ± SD); (c) Colony formation of MDA-MB-231 treated with BAY-299 (0.5, 1, or 2 μM), or DMSO control for 12 days (n = 3); (d) Apoptotic curves of MDA-MB-231 with BAY-299 or DMSO control treatment for 5 days under Annexin V staining. The concentrations of BAY-299 are shown on right (n = 6, mean ± SD); (e) Representative images of MDA-MB-231 with BAY-299 or DMSO treatment for 5 days. The concentrations of BAY-299 are shown on top. The scale bars represent 400 μm; (f) Heatmap showing the Essential scores of *TAF1* and *BRD1* across breast cancer cell lines from the Cancer Dependency Map dataset (https://depmap.org/portal/). Dependency is colored as shown in the bar ranging from red to blue. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

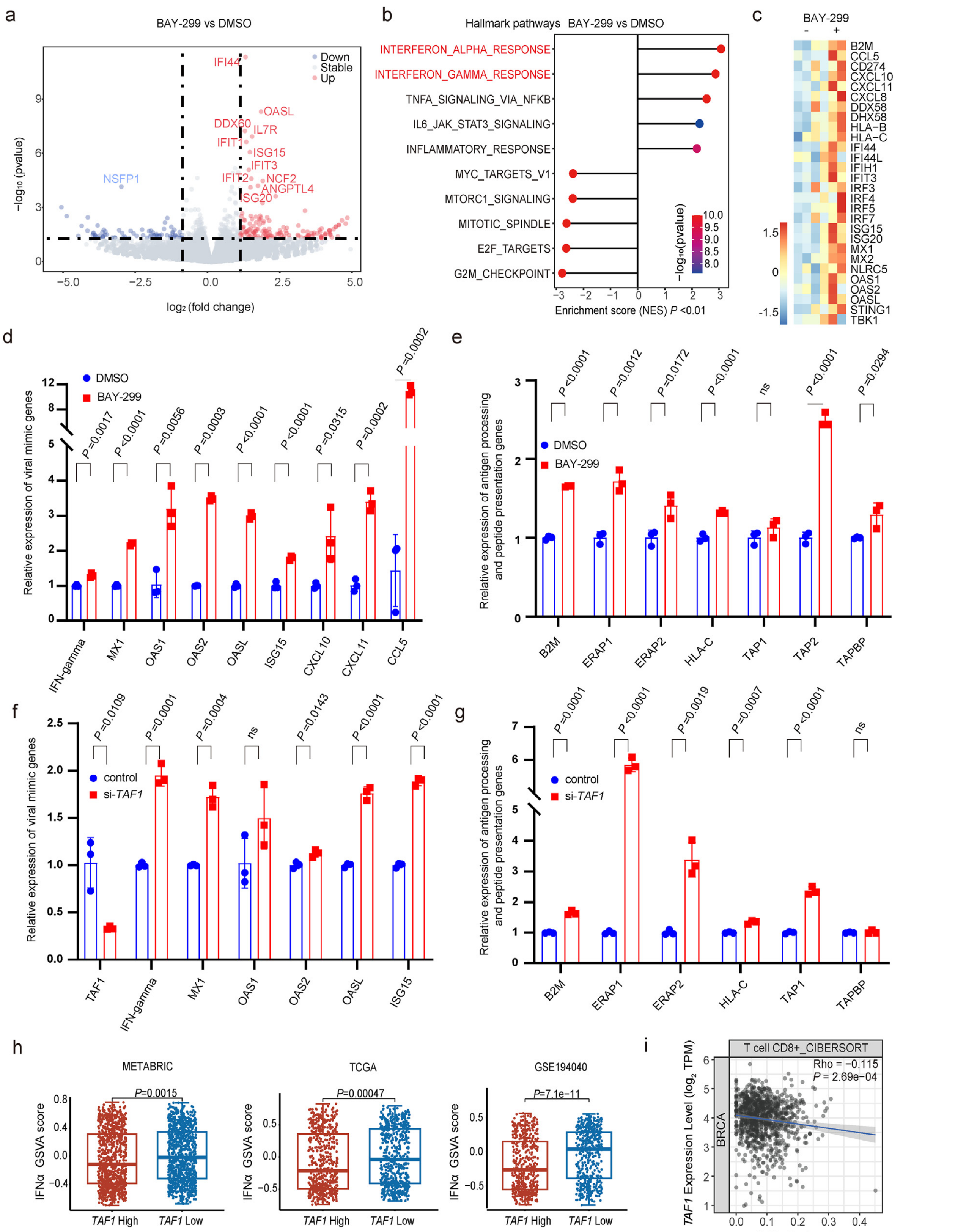

Fig. 2. TAF1 inhibition induces interferon response and enhances antigen processing and presentation in MDA-MB-231 cells. (a) Volcano plot of genes significantly upregulated (red, right) or downregulated (blue, left) following 5 days of BAY-299 treatment in MDA-MB-231 cells (Pearson correlation test); (b) GSEA reveals top ten pathways enriched in MDA-MB-231 after BAY-299 treatment. (NES, normalized enrichment score; one-tailed Fisher's exact test); (c) Heatmap showing RNA-seq data for immune related genes in MDA-MB-231 cells after treated with BAY-299 (+) or DMSO control (−) for 5 days (n = 3); (d) Expression of antiviral signaling genes in MDA-MB-231 cells with DMSO or BAY-299 treatment for 5 days analyzed by qPCR. p-Values are labeled on the top of each pair (n = 3, Student *t* tests); (e) qPCR indicates the induction of antigen processing and presentation genes in MDA-MB-231 cells upon TAF1 inhibition. p-Values are labeled on the top of each pair. ns = not significant (Student t tests); (f) Expression of antiviral signaling genes in *TAF1*-

viral responses following TAF1 inhibition (Fig. 2c–d).

It has been reported that IFNs are essential in upregulating chemokines that attract immune cells, promoting effector T cell recruitment and immune microenvironment reprogramming [18,19]. Correspondingly, TAF1 inhibition caused an increase in the expression of chemokines (*CXCL10*, *CXCL11* and *CCL5*), which are commonly regulated by IFN-gamma proteins (Fig. 2c). The upregulation of all these chemokines was further verified by qPCR (Fig. 2d). Moreover, GSEA also revealed that antigen processing and presentation pathways were significantly enriched upon TAF1 inhibition (NES = 1.4463, $P = 0.01847$, one-tailed Fisher's exact test). The expression of genes associated with antigen processing (*TAP2*) and major histocompatibility complex class I (MHC-I) was enhanced following BAY-299 treatment (Fig. 2e). MHC-I proteins are required for the effective display of antigens to effector T cells [20]. To confirm the on-target effect of TAF1 inhibition, we performed *TAF1* genetic knockdown assay with short interfering RNA (siRNA) in MDA-MB-231 cells. The knockdown of *TAF1* was validated by qPCR measurement (Fig. 2f). As expected, *TAF1* knockdown resulted in an activation of IFN responses as evidenced by a significant increased expression of IFN-gamma, IFN-stimulated genes (Fig. 2f). And the upregulation of genes associated with antigen processing and presentation was observed in *TAF1* knockdown cells (Fig. 2g).

To investigate the clinical significance of TAF1 in anti-tumor immunity, the correlation between *TAF1* expression and interferon alpha pathway was evaluated in breast cancers using three independent clinical datasets (METABRIC, TCGA and GES194040). Gene set variation analysis revealed that tumors with high *TAF1* expression exhibited low GSVA score for the IFN-alpha signature, suggesting that TAF1 repressed innate immune response in breast cancer (Fig. 2h). To further understand the impact of TAF1-associated IFN signatures on tumor microenvironment, we analyzed the correlation between *TAF1* mRNA expression and immune cell infiltrations. Consistent with our findings, CIBERSORT analysis of the TCGA dataset revealed that tumors with high *TAF1* expression levels had lower levels of $CD8^+$ T cells infiltration (Fig. 2i). Collectively, our results suggest that either pharmacologically or genetically targeting TAF1 promotes an effective IFN-driven anti-tumor immune response.

3.3. TAF1 inhibition induces ERV expression and dsRNA formation

We next investigated the source of interferon responses induced by TAF1 inhibition. Growing evidence have demonstrated that reactivated expression of ERVs (endogenous retroviruses) were able to form dsRNA and thus lead to IFN activation [21–24]. Analysis of repetitive element expression revealed that LTRs (ERVs) were induced upon TAF1 inhibition (Fig. 3a). This observation was further validated by qPCR, showing an increased expression of multiple human ERVs, such as *HERVL*, *HERVE*, *HERVF*, and *LTR26E* after TAF1 inhibition (Fig. 3b). Knockdown of *TAF1* also resulted a significant upregulation of ERVs validated by qPCR (Fig. 3c).

To investigate whether these ERVs transcription leads to the formation of dsRNA, immunofluorescence staining with J2 antibody, which specifically recognizes dsRNA [25], was performed. A significant accumulation of dsRNA was observed in MDA-MB-231 cells treated with BAY-299, suggesting the formation of dsRNA in response to TAF1 inhibition (Fig. 3d). Knockdown of *TAF1* also resulted a significant accumulation of dsRNA (Fig. 3e). Intracellular dsRNA formation was reported to be recognized by cytosolic RNA sensors, MDA5 (*IFIH1*) and RIG-I (*DDX58*) [26], and subsequently resulted in the MAVs activation, which induced downstream antiviral signaling and apoptosis [27]. In agreement, mRNA level of MDA5, RIG-I, and MAVs was significantly increased after BAY-299 treatment or *TAF1* knockdown (Fig. 3f–g). Supporting these findings, gene sets of cellular response to the virus and dsRNA-sensing signaling (RIG-I signaling pathway) were significantly enriched upon TAF1 inhibition (Fig. 3h). Collectively, our data revealed that TAF1 inhibition triggered ERV expression and dsRNA accumulation, which can be recognized by cytosolic RNA sensors (MDA5 and RIG-I) and thus activates the interferon responses.

3.4. PCNA expression level correlates with BAY-299 sensitivity and the immunity

Given the heterogeneity of TNBC [3], we next asked if TNBC cells have different response to BAY-299 treatment. To address this question, we measured the anti-proliferative effect of BAY-299 across twelve different TNBC cell lines. Serial concentration of BAY-299, ranging from 0.02 to 10 μM, were added to TNBC cells and cellular response was determined by AACs (area above curve) after 120 h treatment (Fig. 4a). We observed the heterogeneous growth inhibitory effects across these TNBC cell lines. HCC1395 was the most sensitive cell line with $IC_{50} = 0.4044$ μM, while BT-549 was the most insensitive cell line with $IC_{50} > 10$ μM, the maximum dose used in this experiment (Fig. 4b).

In order to investigate the molecular biomarker that could predict the response of TNBC to BAY-299 treatment, we performed a correlation analysis between BAY-299 efficacy as measured by AAC and basal gene expression levels in 12 TNBC cell lines. Pearson correlation coefficients were calculated and a rank-ordered gene list was generated for GSEA (Fig. 4d). There were 920 significant genes identified, including 691 upregulated genes and 229 downregulated genes ($P < 0.05$, Fig. 4e). The IFN-alpha pathway was enriched in the sensitive cell lines suggesting that pre-existing IFN response could serve as a potential signature to differentiate the sensitivity of BAY-299 (Fig. 4f). On the other hand, the E2F targets pathway was the most significantly enriched pathway in insensitive cells, suggesting that elevated expression of genes involved in this pathway correlated with the resistance to TAF1 inhibition (Fig. 4f). Moreover, genes associated with the MYC pathway were also enriched in BAY-299 resistant cell lines, suggesting that rapidly proliferating cells are more likely to resist TAF1 inhibition [28].

We next sought to identify a specific protein associated with the responsiveness of TNBC cells to BAY-299. We calculated Pearson correlation coefficients between AAC and protein expression, leading to the identification of 11 significantly correlated proteins [29] (Fig. 4g). Interestingly, only protein proliferating cell nuclear antigen (PCNA) was significantly correlated with BAY-299 response at both mRNA level ($R = -0.680$, $P = 0.014$) and protein level ($R = -0.580$, $P = 0.047$) (Fig. 4i–j). PCNA is a key regulator of DNA replication and cell cycle regulation, validating the correlation between cell proliferating status and TAF1 dependency in TNBC cells.

In light of the regulatory role of TAF1 in TNBC's innate immune responses, we next investigated the possibility of a correlation between *PCNA* and tumor immunity. As expected, a significant negative correlation was observed between *PCNA* levels and NKT cell infiltration in TNBC ($R = -0.269$, $P < 0.0001$) (Fig. 4k). This correlation was also recapitulated in other cancer types, including

knockdown MDA-MB-231 cells analyzed by qPCR (n = 3, Student *t* tests); (g) qPCR indicates the induction of antigen processing and presentation genes in *TAF1*-knockdown MDA-MB-231 cells (n = 3, Student *t* tests); (h) GSVA scores of interferon alpha signature in METABRIC, TCGA and GES194040. p-Values were labeled on the top of each pair (Student *t* tests); (i) The correlation of *TAF1* and the $CD8^+$ T cell infiltration in BRCA analyzed via CIBERSORT from TCGA. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

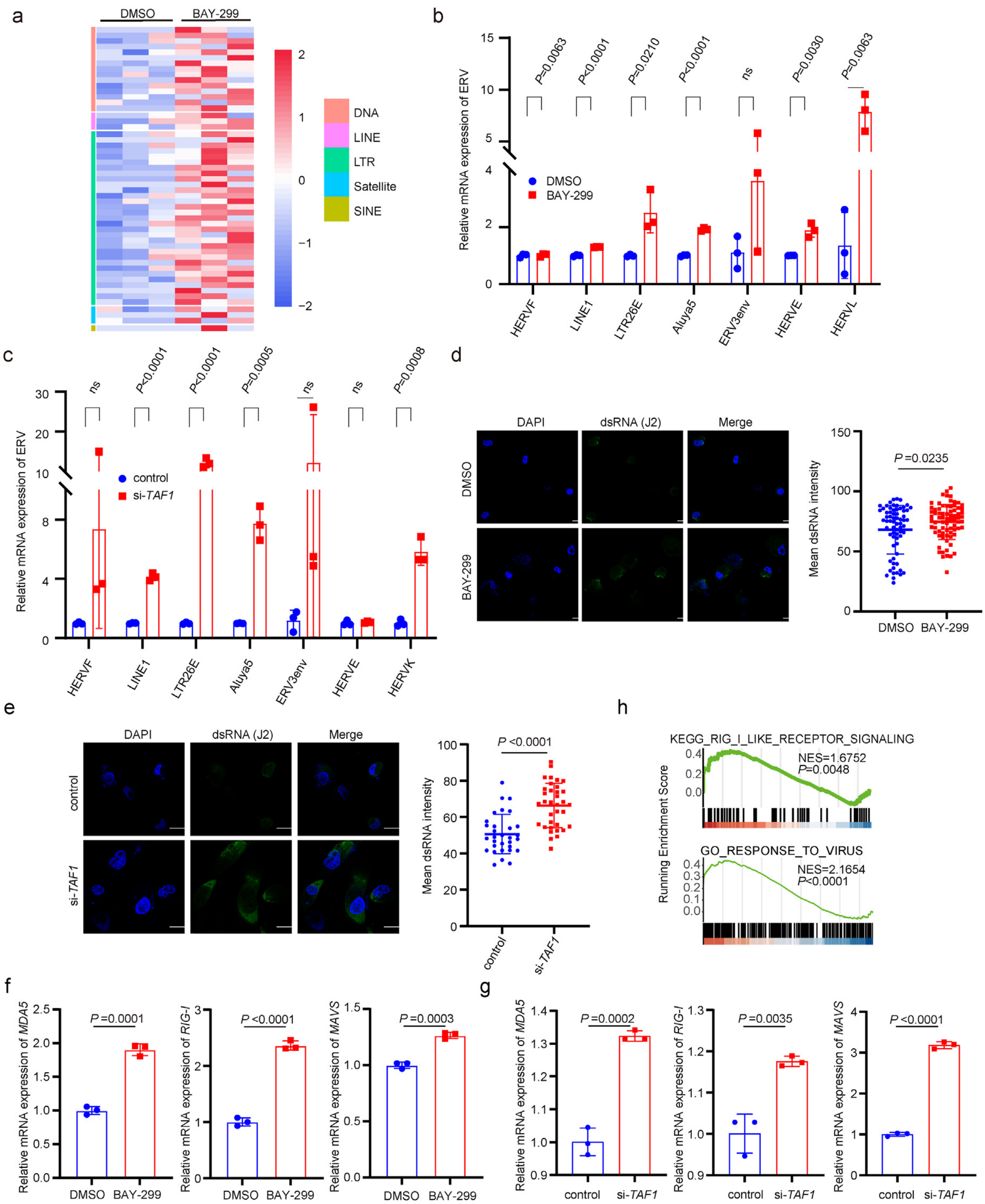

Fig. 3. TAF1 inhibition induces the expression of ERVs and dsRNA formation in MDA-MB-231 cells. (a) Heatmap shows the differential expression of repetitive elements upon TAF1 inhibition, accessed by RNA-seq. Down-regulated expression are shown in blue and up-regulated expressions are shown in red (n = 3); (b) ERVs are induced upon BAY-299 treatment in MDA-MB-231 cells assessed by qPCR. p-Values are labeled on the top of each pair (n = 3, Student *t* tests); (c) ERVs are induced in *TAF1*-knockdown MDA-MB-231 cells assessed by qPCR. p-Values are labeled on the top of each pair (n = 3, Student *t* tests); (d) Immunofluorescent detection of dsRNA in MDA-MB-231 induced upon TAF1 inhibition using dsRNA specific J2 antibody (Student *t* tests). The scale bars represent 20 μm; (e) Immunofluorescent detection of dsRNA in *TAF1*-knockdown MDA-MB-231 cells using dsRNA specific J2 antibody (Student t tests). The scale bars represent 20 μm; (f) The dsRNA receptors induced upon TAF1 inhibition in MDA-MB-231 cells were assessed by qPCR (n = 3, Student *t* tests); (g) The dsRNA receptors induced in *TAF1*-knockdown MDA-MB-231 cells were assessed by qPCR (n = 3, Student *t* tests); (h) GSEA of RIG-I like receptor signaling (NES = 1.6752, *P* = 0.0048) and response to virus pathways (NES = 2.1654, *P* < 0.0001) between BAY-299 treated cells and DMSO control (One-tailed Fisher's exact test). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

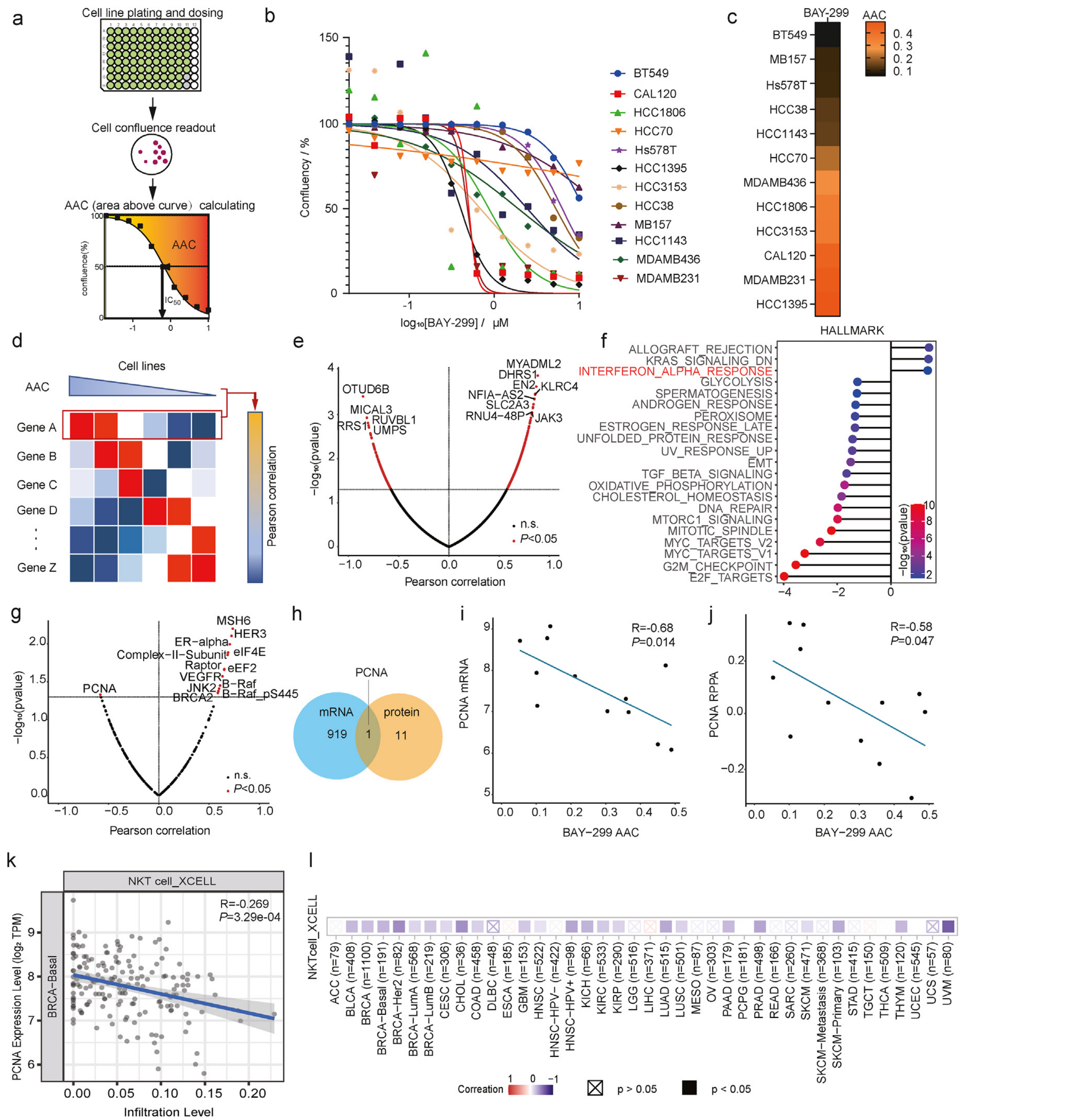

Fig. 4. Heterogeneous response of TNBC cell lines to BAY-299. (a) Schematic graph of cell line sensitivity calculation by AACs; (b) Growth curves of 12 TNBC cell lines after BAY-299 treatment with indicated concentrations for 5 days. Cell lines are labeled on the right (n = 4); (c) Heatmap of BAY-299 responsiveness in 12 TNBC cell lines. AACs were calculated from the dose-response curves. Data are normalized to DMSO. Area above curve differentials are colored as shown in the bar ranging from blank to orange; (d) Schematic graph of biomarker identification by correlating AAC to gene expression; (e) Pearson's correlation coefficients between log_2 normalized mRNA expression data and response to BAY-299. Significantly regulated genes are colored in red dots, and the top regulated genes are labeled (Student *t* tests); (f) Top enriched pathways in BAY-299 sensitive cell lines compared to resistant cell lines based on GSEA. The interferon alpha response pathway is labeled in red; (g) Volcano plot showing proteins significantly associated with sensitive cell lines (left) or resistant cell lines (right) based on Princess Margaret Cancer Center (PMCC) data. Significantly regulated genes are colored in red, and the top regulated genes are labeled (Student *t* tests); (h) Venn diagram illustrating the overlap between mRNA and protein significantly associated with sensitivity; (i) Significant correlation between *PCNA* mRNA level and BAY-299 sensitivity. ($R = -0.680$, $P = 0.014$, Student *t* tests); (j) Significant correlation between *PCNA* protein level and BAY-299 sensitivity. ($R = -0.580$, $P = 0.047$, Student *t* tests); (k) CIBERSORT analysis of the correlation between *PCNA* and the NKT cell infiltration in TNBC ($R = -0.259$, $P = 3.29e-04$); (l) Heatmap showing the negative correlation between *PCNA* mRNA levels and NKT cells infiltration across multiple cancers. Correlation is colored as shown in the bar ranging from blue to red. Red: positive correlation ($P < 0.05$, $r > 0$), blue: negative correlation ($P < 0.05$, $r < 0$), x: not significant ($P > 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

uveal melanoma and cholangiocarcinoma (Fig. 4l). These findings are consistent with previous studies showing that PCNA overexpression inhibits the expression of interferon [30].

In summary, our study suggested that the high expression level of PCNA may serve as a marker of suppressive tumor immune responses, which could limit the efficacy of TAF1 inhibitor in various cancers.

4. Discussion

TAF1 has emerged as a promising therapeutic target in different cancer types. Inhibition of TAF1 induces cell cycle arrest and apoptosis, leading to tumor suppression. In this study, we discovered a novel role of TAF1 in regulating innate antitumor immunity. TAF1 inhibition was found to enhance the antitumoral immune response via multiple mechanisms, including the upregulation of ERVs expression and dsRNA production, ultimately leading to the activation of interferon responses, cytokine production, and antigen processing and presentation in TNBC. Consistent with our findings, the analysis of public clinical datasets from breast cancer patients revealed a negative correlation between *TAF1* expression and IFN-alpha signature. These results support the growing evidence that TAF1 inhibitors may improve the efficacy of immune checkpoint blockade (ICB) treatment. However, further studies are required to fully understand the effects of TAF1 inhibition on immune cells.

Due to the heterogeneity of TNBC, there is a need to identify predictive biomarkers for drug response. Our gene expression data revealed that TNBC cells with elevated PCNA expression are more resistant to TAF1 inhibition by BAY-299. However, further investigations are required to uncovered whether PCNA inhibition with small molecular inhibitors such as T2AA could sensitize resistant cells to BAY-299 [31].

GSEA analysis revealed two HALLMARK terms, E2F targets and G2M checkpoint, which are significantly downregulated with BAY-299 treatment, implying that TAF1 inhibition disrupts cell cycle progression. Emerging evidence suggests that cell cycle disruption concurrently triggers immune response signaling, thereby activating a robust immune surveillance program mediated by T cells, and further enhanced by ICB therapy [32]. Based on our findings, we propose that the combination of TAF1 inhibitors with CDK inhibitors can sensitize ICB therapy by restoring IFN responses, making this combination an attractive therapeutic strategy against refractory TNBC.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Triple negative breast cancer: emerging therapeutic modalities and novel combination therapies, Cancer Treat Rev. 62 (2018) 110–122.
[2] J. Collignon, L. Lousberg, H. Schroeder, G. Jerusalem, Triple-negative breast cancer: treatment challenges and solutions, Breast Cancer 8 (2016) 93–107.
[3] G. Bianchini, J.M. Balko, I.A. Mayer, M.E. Sanders, L. Gianni, Triple-negative breast cancer: challenges and opportunities of a heterogeneous disease, Nat. Rev. Clin. Oncol. 13 (2016) 674–690.
[4] H. Yagata, Y. Kajiura, H.J.B.C. Yamauchi, Current strategy for triple-negative breast cancer: appropriate combination of surgery, radiation, and chemotherapy, Breast Cancer 18 (2011) 165–173.
[5] M.A. Clegg, N.H. Theodoulou, P. Bamborough, C.W. Chung, P.D. Craggs, E.H. Demont, L.J. Gordon, G.M. Liwicki, A. Phillipou, N.C.O. Tomkinson, et al., Optimization of naphthyridones into selective TATA-binding protein associated factor 1 (TAF1) bromodomain inhibitors, ACS Med. Chem. Lett. 12 (2021) 1308–1317.
[6] R.M. Karim, L. Yang, L. Chen, M.J. Bikowitz, J. Lu, D. Grassie, Z.P. Shultz, J.M. Lopchuk, J. Chen, E. Schonbrunn, Discovery of dual TAF1-ATR inhibitors and ligand-induced structural changes of the TAF1 tandem bromodomain, J. Med. Chem. 65 (2022) 4182–4200.
[7] N. Tanese, B.F. Pugh, R. Tjian, Coactivators for a proline-rich activator purified from the multisubunit human TFIID complex, Genes Dev. 5 (1991) 2212–2224.
[8] C.A. Mizzen, The TAF(II)250 subunit of TFIID has histone acetyltransferase activity, Cell 87 (1996) 1261–1270.
[9] S.L. Kloet, J.L. Whiting, P. Gafken, J. Ranish, E.H. Wang, Phosphorylation-dependent regulation of cyclin D1 and cyclin A gene transcription by TFIID subunits TAF1 and TAF7, Mol. Cell Biol. 32 (2012) 3358–3369.
[10] Q. Li, Q. Wu, Z. Li, Y. Hu, F. Zhou, Z. Zhai, S. Yue, H. Tian, LncRNA LINC00319 is associated with tumorigenesis and poor prognosis in glioma, Eur. J. Pharmacol. 861 (2019), 172556.
[11] J. Zhang, R. Li, B. Zhang, X. Cui, TAF1 promotes NSCLC cell epithelial-mesenchymal transition by transcriptionally activating TGFbeta1, Biochem. Biophys. Res. Commun. 636 (2022) 113–118.
[12] P. Tavassoli, L.A. Wafa, H. Cheng, A. Zoubeidi, L. Fazli, M. Gleave, R. Snoek, P.S. Rennie, TAF1 differentially enhances androgen receptor transcriptional activity via its N-terminal kinase and ubiquitin-activating and -conjugating domains, Mol. Endocrinol. 24 (2010) 696–708.
[13] L. Zhou, Q. Yao, L. Ma, H. Li, J. Chen, TAF1 inhibitor Bay-299 induces cell death in acute myeloid leukemia, Transl. Cancer Res. 10 (2021) 5307–5318.
[14] T. Li, J. Fan, B. Wang, N. Traugh, Q. Chen, J.S. Liu, B. Li, X.S. Liu, TIMER: a web server for comprehensive analysis of tumor-infiltrating immune cells, Cancer Res. 77 (2017) e108.
[15] R. Marcotte, A. Sayad, K.R. Brown, F. Sanchez-Garcia, J. Reimand, M. Haider, C. Virtanen, J.E. Bradner, G.D. Bader, G.B.J.C. Mills, Functional genomic landscape of human breast cancer drivers, vulnerabilities, and resistance, Cell 164 (2016) 293–309.
[16] L. Bouche, C.D. Christ, S. Siegel, A.E. Fernandez-Montalvan, S.J. Holton, O. Fedorov, A. Ter Laak, T. Sugawara, D. Stockigt, C. Tallant, et al., Benzoisoquinolinediones as potent and selective inhibitors of BRPF2 and TAF1/TAF1L bromodomains, J. Med. Chem. 60 (2017) 4002–4022.
[17] A. Tsherniak, F. Vazquez, P.G. Montgomery, B.A. Weir, G. Kryukov, G.S. Cowley, S. Gill, W.F. Harrington, S. Pantel, J.M. Krill-Burger, et al., Defining a cancer dependency Map, Cell 170 (2017) 564–576, e516.
[18] J. Li, K.T. Byrne, F. Yan, T. Yamazoe, Z. Chen, T. Baslan, L.P. Richman, J.H. Lin, Y.H. Sun, A.J. Rech, et al., Tumor cell-intrinsic factors underlie heterogeneity of immune cell infiltration and response to immunotherapy, Immunity 49 (2018) 178–193, e177.
[19] A.S. Dighe, E. Richards, L.J. Old, R.D.J.I. Schreiber, Enhanced in vivo growth and resistance to rejection of tumor cells expressing dominant negative IFN gamma receptors, Immunity 1 (1994) 447–456.
[20] K. Dhatchinamoorthy, J.D. Colbert, K.L. Rock, Cancer immune evasion through loss of MHC class I antigen presentation, Front. Immunol. 12 (2021), 636568.
[21] I. Canadas, R. Thummalapalli, J.W. Kim, S. Kitajima, R.W. Jenkins, C.L. Christensen, M. Campisi, Y. Kuang, Y. Zhang, E. Gjini, et al., Tumor innate immunity primed by specific interferon-stimulated endogenous retroviruses, Nat. Med. 24 (2018) 1143–1150.
[22] D. Roulois, H. Loo Yau, R. Singhania, Y. Wang, A. Danesh, S.Y. Shen, H. Han, G. Liang, P.A. Jones, T.J. Pugh, et al., DNA-demethylating agents target colorectal cancer cells by inducing viral mimicry by endogenous transcripts, Cell 162 (2015) 961–973.
[23] K.B. Chiappinelli, P.L. Strissel, A. Desrichard, H. Li, C. Henke, B. Akman, A. Hein, N.S. Rote, L.M. Cope, A. Snyder, et al., Inhibiting DNA methylation causes an interferon response in cancer via dsRNA including endogenous retroviruses, Cell 162 (2015) 974–986.
[24] W. Sheng, M.W. LaFleur, T.H. Nguyen, S. Chen, A. Chakravarthy, J.R. Conway, Y. Li, H. Chen, H. Yang, P.H. Hsu, et al., LSD1 ablation stimulates anti-tumor immunity and enables checkpoint blockade, Cell 174 (2018) 549–563, e519.
[25] J. Schonborn, J. Oberstraβ, E. Breyel, J. Tittgen, J. Schumacher, N.J. Lukacs, Monoclonal antibodies to double-stranded RNA as probes of RNA structure in crude nucleic acid extracts, Nucleic Acids Res. (1991) 2993–3000.
[26] J. Rehwinkel, M.U. Gack, RIG-I-like receptors: their regulation and roles in RNA sensing, Nat. Rev. Immunol. 20 (2020) 537–551.
[27] S. El Maadidi, L. Faletti, B. Berg, C. Wenzl, K. Wieland, Z.J. Chen, U. Maurer, C.J. Borner, A novel mitochondrial MAVS/Caspase-8 platform links RNA virus-induced innate antiviral signaling to Bax/Bak-independent apoptosis, J. Immunol. 192 (2014) 1171–1183.
[28] G. Bretones, M. Delgado, J.L. Mechanisms, Myc and cell cycle control, Biochim. Biophys. Acta 1849 (2015) 506–516.
[29] R. Marcotte, A. Sayad, K.R. Brown, F. Sanchez-Garcia, J. Reimand, M. Haider, C. Virtanen, J.E. Bradner, G.D. Bader, G.B. Mills, et al., Functional genomic

landscape of human breast cancer drivers, vulnerabilities, and resistance, Cell 164 (2016) 293–309.
[30] H. Zhang, Y. Chen, J. Cui, X. Yan, Y. Sun, T.J.F. Xu, S. Immunology, PCNA negatively regulates MITA through the autophagy pathway in miiuy croaker (Miichthys miiuy), Fish Shellfish Immunol. 124 (2022) 21–27.
[31] C. Punchihewa, A. Inoue, A. Hishiki, Y. Fujikawa, M. Connelly, B. Evison, Y. Shao, R. Heath, I. Kuraoka, P.J. Rodrigues, Identification of a small molecule PCNA inhibitor that disrupts interactions with PIP-Box proteins and inhibits DNA replication, J. Biol. Chem. 63 (2012) 1021–1027.
[32] S. Goel, M.J. DeCristo, A.C. Watt, H. BrinJones, J. Sceneay, B.B. Li, N. Khan, J.M. Ubellacker, S. Xie, O. Metzger-Filho, et al., CDK4/6 inhibition triggers anti-tumour immunity, Nature 548 (2017) 471–475.